MILITARY SPECIFICATION SHEET

SWITCHES, PRESSURE, (GAGE) TYPE II,
~~LOW LEVEL TO 5 AMPERES~~

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense

The complete requirements for procuring the pressure switches described herein shall consist of this document and the latest issue of Specification MIL-S-9395

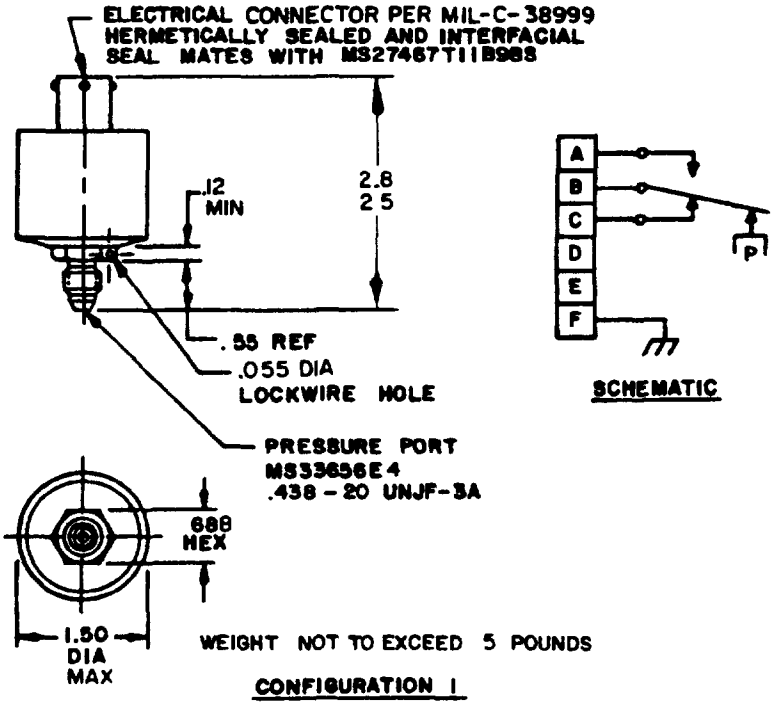
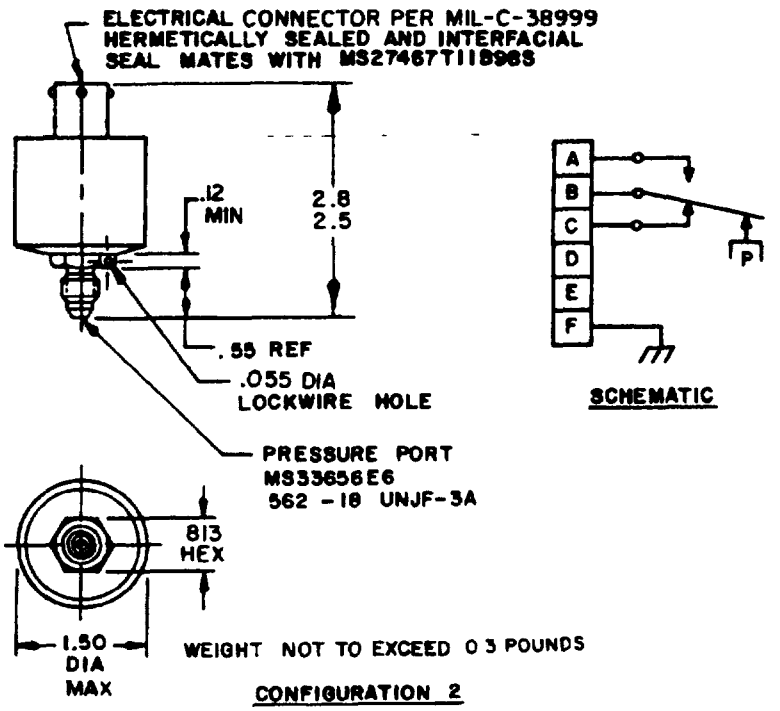


FIGURE 1 Switches

FSC 5930

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INCHES	MM
055	1 40
12	3.0
55	14 0
688	17 48
813	20 65
2 5	64
2 8	71

NOTES
1 Dimensions are in inches
2 Metric equivalents are given for general information only and are based upon 1 00 inch = 25 4 mm
3 Exact shape of switch is optional provided outline dimensions specified are not exceeded and connector locations are as specified
4 Schematics shown are for switches with pressure ports exposed to zero lb_f/in^2

FIGURE 1. Switches - Continued

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REQUIREMENTS

Dimensions, configuration, and electrical schematic See figure 1

Weight See figure 1

Calibration See tables I, II, III, and IV

System pressure 125 lb_f/in^2

Proof pressure 250 lb_f/in^2

Burst pressure 500 lb_f/in^2

Electrical ratings See table I
Minimum current 25,000 cycles.
Low level 50,000 cycles
NOTE Switches shall be subjected only to low level loads prior to delivery

Seal
Electrical chamber See table I
Pressure chamber Hermetic
Reference chamber Unsealed

Electrical connector See figure 1

Pressure port See figure 1

Media Dry air, nitrogen gas, fuel per MIL-T-5624, lubricating oil per MIL-L-7808, hydraulic fluid per MIL-H-5606 or MIL-H-83282, oxygen, or Coolanol 25R, or equal

High temperature (operating and nonoperating) D (400°F and 500°F for 30 minutes)

Low temperature (operating and nonoperating) D (-65°F)

Altitude C (except 80,000 ft)

Shock C (100 G)

Vibration S (test condition D, method 204 of MIL-STD-202, except 10 to 2,000 Hz, 20 G)

Life (mechanical) A (100,000 cycles)

Life (electrical) C (50,000 cycles)

Acceleration C (8 G)

Pulsation amplitude A (0%)

Pulsation frequency A (0 Hz)

Pressure rise A (less than 100 lb_f/in^2)

Dielectric withstanding voltage (at reduced barometric pressure) Applicable at 350 Vrms

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Electrical connector torque 8 foot-pounds

Switch mounting torque 15 foot-pounds

Terminal strength Applicable

Flame test Applicable

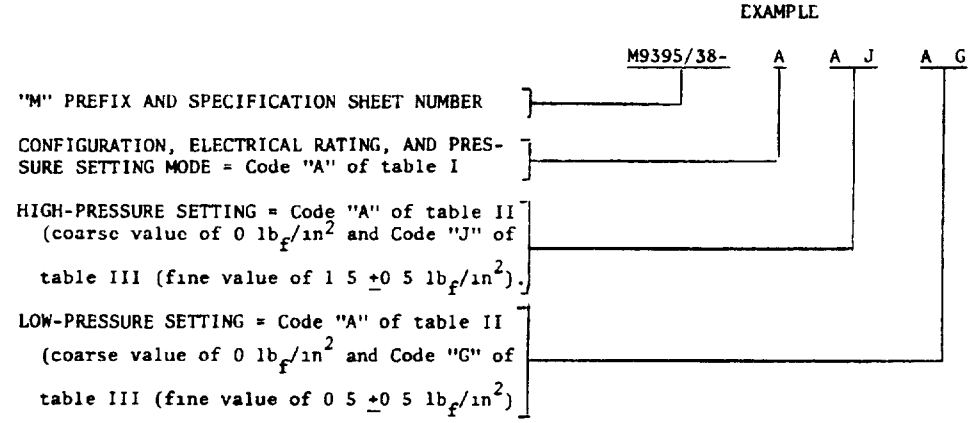
Explosion Applicable

QUALIFICATION

Single submission Restricted to switch submitted
Group submission See table V

PART NUMBER Consists of "M" prefix followed by specification sheet number, a dash (-), and a five-letter code The five-letter code identifies the configuration, electrical rating, and pressure setting mode (code from table I), high-pressure setting (coarse value code from table II) followed by fine value with applicable tolerance (code from table III), and low-pressure setting (coarse value code from table II) followed by fine value with applicable tolerance (code from table III) The five-letter code used in the following example identifies a switch of configuration 1, low level to 1 ampere resistive at 28 Vdc, which actuates on increasing pressure at 1 5 $\pm$0 5 lb_f/in^2 and deactuates on decreasing pressure at 0 5 $\pm$0 5 lb_f/in^2

EXAMPLE



NOTE Design limitations (actuation values and tolerances, deadband and deactuation values and tolerances) should be coordinated with manufacturer(s) listed on the QPL for this specification sheet before specifying a particular "M" number The fact that operating characteristics can be coded does not necessarily mean that it can be manufactured or procured

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TABLE I. Codes for combinations of configurations, electrical ratings, and pressure setting modes

	Low level to 1 A resistive at 28 Vdc				1.5 to 5 A resistive at 28 Vdc				Pressure setting mode	
	Configuration				Configuration					
	1		2		1		2			
	Electrical chamber		Electrical chamber		Electrical chamber		Electrical chamber		High setting	Low setting
	Hermetic	Unsealed	Hermetic	Unsealed	Hermetic	Unsealed	Hermetic	Unsealed		
Code -	A	D	G	K	N	R	U	X	At (or max) 1/	At (or min) 1/
Code -	B	E	H	L	P	S	V	Y	At (or max) 1/	Differential 2/
Code -	C	F	J	M	Q	T	W	Z	Differential 2/	At (or min) 1/

1/ Setting values are designated by codes from tables II and III

2/ Setting values are designated by codes from table IV

TABLE II. Codes for coarse settings

Code	Coarse value (lb_f/in^2)	Code	Coarse value (lb_f/in^2)	Code	Coarse value (lb_f/in^2)
A - - -	0	L - - -	30	W - - -	80
B - - -	2 5	M - - -	35	X - - -	85
C - - -	5	N - - -	40	Y - - -	90
D - - -	7 5	P - - -	45	Z - - -	95
E - - -	10	Q - - -	50	1 - - -	100
F - - -	12 5	R - - -	55	2 - - -	105
G - - -	15	S - - -	60	3 - - -	110
H - - -	17.5	T - - -	65	4 - - -	115
J - - -	20	U - - -	70	5 - - -	120
K - - -	25	V - - -	75	6 - - -	125

TABLE III Codes for combinations of fine settings and tolerance values

	Fine value (lb_f/in^2) for settings below 20 lb_f/in^2					Tolerance (lb_f/in^2)
	0	0.5	1	1.5	2	
Code - - -	A	B	C	D	E	±0.25
Code - - -	F	G	H	J	K	±0 5
Code - - -	L	M	N	P	Q	±1.0
Code - - -	R	S	T	U	V	±1 5
Code - - -	W	X	Y	Z	1	±2 0
Code - - -	2	3	4	5	6	±2.5
Code - - -	7	8	9	0	-	Min or Max
	Fine value (lb_f/in^2) for settings of 20 lb_f/in^2 and above					Tolerance (lb_f/in^2)
	0	1	2	3	4	
Code - - -	A	B	C	D	E	±1.0 1/
Code - - -	F	G	H	J	K	±2 0 2/
Code - - -	L	M	N	P	Q	±3 0 3/
Code - - -	R	S	T	U	V	±4 0
Code - - -	W	X	Y	Z	1	±5 0
Code - - -	2	3	4	5	6	±6 0
Code - - -	7	8	9	0	-	Min or Max

1/ Not applicable for pressure settings above 33 lb_f/in^2

2/ Not applicable for pressure settings above 66 lb_f/in^2

3/ Not applicable for pressure settings above 100 lb_f/in^2

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TABLE IV Codes for differential settings 1/

Code	Differential value (lb_f/in^2)	Code	Differential value (lb_f/in^2)
A - - -	0	T - - -	11
B - - -	0 5	U - - -	12
C - - -	1	V - - -	13
D - - -	1 5	W - - -	14
E - - -	2	X - - -	15
F - - -	2 5	Y - - -	16
G - - -	3	Z - - -	18
H - - -	3 5	1 - - -	20
J - - -	4	2 - - -	22
K - - -	4.5	3 - - -	24
L - - -	5	4 - - -	26
M - - -	5 5	5 - - -	28
N - - -	6	6 - - -	30
P - - -	7	7 - - -	35
Q - - -	8	8 - - -	40
R - - -	9	9 - - -	45
S - - -	10	0 - - -	50

1/ Differential settings require two codes, minimum differential and maximum differential

TABLE V Extent of qualification

Part number	No of samples required	Tests	Qualifies
ELECTRICAL CHAMBERS EITHER HERMETICALLY SEALED OR UNSEALED			
M9395/38-AAEAH -NAEAH -N6R4R -A6R4R	2 ea. resistive 2 ea intmd current 2 ea. low level 2 ea resistive	Complete per qualification inspection on MIL-S-9395	All
ELECTRICAL CHAMBERS UNSEALED			
M9395/38-BAEAH -DAEAH -D6R4R -X6R4R	2 ea. resistive 2 ea intmd current 2 ea. low level 2 ea. resistive	Complete per qualification inspection of MIL-S-9395	Configuration codes D, E, F, K, L, M, R, S, T, X, Y, and Z

NOTE Revision letters are not used to denote changes due to the extensiveness of the changes

Custodians
Army - EL
Navy - EC
Air Force - 85

Preparing activity
Air Force - 85

(Project 5930-1055-4)

Review activities
Army - AV
Air Force - 11, 17, 99
DLA - ES

User activities
Army - AT
Navy - MC, AS, SH
Air Force - 19

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MIL-S-9395/38A Switches, Pressure (Gage) Type II, Low Level to 5 Amperes FSC 5930

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